Optimization of Laser Keyhole Welding Strategies of Dissimilar Metals by FEM Simulation

VIRGINIA GARCIA NAVAS, JOSU LEUNDA, JON LAMBARRI, and CARMEN SANZ

Laser keyhole welding of dissimilar metals has been simulated to study the effect of welding strategies (laser beam displacements and tilts) and combination of metals to be welded on final quality of the joints. Molten pool geometry and welding penetration have been studied but special attention has been paid to final joint material properties, such as microstructure/phases and hardness, and especially to the residual stress state because it greatly conditions the service life of laser-welded components. For a fixed strategy (laser beam perpendicular to the joint) austenitic to carbon steel laser welding leads to residual stresses at the joint area very similar to those obtained in austenitic to martensitic steel welding, but welding of steel to Inconel 718 results in steeper residual stress gradients and higher area at the joint with detrimental tensile stresses. Therefore, when the difference in thermo-mechanical properties of the metals to be welded is higher, the stress state generated is more detrimental for the service life of the component, and consequently more relevant is the optimization of welding strategy. In laser keyhole welding of austenitic to martensitic stainless steel and austenitic to carbon steel, the optimum welding strategy is displacing the laser beam 1 mm toward the austenitic steel. In the case of austenitic steel to Inconel welding, the optimum welding strategy consists in setting the heat source tilted 45 deg and moved 2 mm toward the austenitic steel.

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I. INTRODUCTION

NOWADAYS, joining dissimilar metals is indispensable in manufacturing and constructing advanced equipment and machinery. Different kinds of metals feature different chemical, physical, and metallurgical properties: some are more resistant to corrosion, some are lighter, some are stronger, *etc*. The aim of joining dissimilar metals is, therefore, to achieve different properties of metals in different parts of a component. This maximizes the performance of the component minimizing material costs. For dissimilar metal welds, it is common practice that mechanical properties of the joint should not be worse than those of the inferior base metal.

Different methods of joining metals can be used: fusion welding, pressure welding, explosion welding, friction welding, diffusion welding, brazing, *etc*. Laser keyhole welding is one of the most promising joining methods because, compared with conventional welding techniques, it provides high productivity (associated to the high welding speeds), narrow joint and heat-affected zone (HAZ), manufacturing flexibility, and ease of automation.

Dissimilar metal laser-weld failures occur in many industrial applications, generally attributed to the different metallurgical and physical properties between the two materials, such as laser wavelength absorption, thermal conductivity, reflectivity, melting point, possible formation of brittle phases, and wettability. These different material properties, and as a consequence the different material response to laser, are the origin of the high difficulties of laser-welding dissimilar metals.[1] Consequently, much work should be done to define parameter-operating windows for the different alloys of industrial interest to produce laser-welded joints of dissimilar metals free of defects and with good material properties to guarantee the good performance of the welded component.

The effect of laser parameters (laser power, welding speed, defocus distance, laser pulse energy, duration and repetition rates in pulsed lasers, laser beam diameter, *etc*.) on the final characteristics of dissimilar laser welds can be found in several experimental studies.[2–19] Most of these works are focused on the effect of laser parameters on weld bead geometry although some of them study also other characteristics of the joints such as the mixing behavior of the materials in the fusion zone, the microstructure, the presence of defects, hardness, and mechanical properties. Some authors[20–24] state that a displacement and/or tilting of the laser beam is a good strategy for welding dissimilar materials, able to yield good mechanical properties, high reliability, and repeatability in an industrial application. Panton *et al.*[5] state also that positioning of the laser beam greatly affects hardness and susceptibility to cracking in the fusion zone, which heavily impact the weld strength. Therefore, as is also pointed by Khan *et al.*,[6,7] beam incident angle and position are key factors that

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determine the weld characteristics in laser beam welding of dissimilar metals. To define the optimum laser-welding strategy in each particular case (size and geometry of components and materials of the parts to be welded), a high number of experimental trial and error tests would be necessary, turning out time consuming and implying elevated costs. Simulation of the welding process can be employed to predict weld attributes and thus can partially replace the expensive, time-consuming experimental-based trial and error method in the development of new welding procedures.[25] Thus, industrially, it is essential to have simulation tools that permit the optimization of processing strategies in each particular case.

Mackwood and Crafer[1] and Abderrazak *et al.*[26] carried out extended reviews of the different analytical and numerical solutions commonly employed in the simulation of laser welding. In the literature, numerous works[26–37] related to simulation of laser keyhole welding can be found, but most of them are focused only on the study of the thermal fields and weld pool geometry (bead length and width and depth of penetration). In fact, few or none of these works focus their studies on modeling microstructure, hardness, and, more important, the residual stresses generated in laser welding of dissimilar metals. Residual stress in welding occurs due to misfit between different parts, different phases, or different regions within the same part arising out of non-uniform thermal strain, and strains arising from solidification and solid state phase transformations.[38] Residual stress state is a key characteristic of the welded joints: residual stresses add up to the external stresses that the component suffers under working conditions and therefore widely condition the service life of laser-welded components. Missori and Koerber[39] state that problems related to the use of dissimilar metal welds are associated with premature failures during service, connected in part to thermal stresses generated at the weld interface: welding-induced tensile residual stresses promote brittle fracture, buckling deformation, and stress-corrosion cracking and reduce the fatigue life of welded structures in service.[38]

To fill this gap observed in the literature, in this work laser keyhole welding of dissimilar metals has been simulated to study the effect of different welding strategies (displacement and tilt of the laser beam) on the final quality of the welds. Special attention has been paid to the final residual stresses in the weld, but geometry of molten pool, welding penetration, microstructure, and hardness have been also studied. Some experimental results (hardness, microstructure, and residual stresses) have been used to validate the simulations.

II. EXPERIMENTAL PROCEDURE

A. *Geometry and Materials*

In the present work, laser keyhole welding of two cylindrical parts (Figure 1) has been simulated. One of the parts is shorter than the other just to distinguish

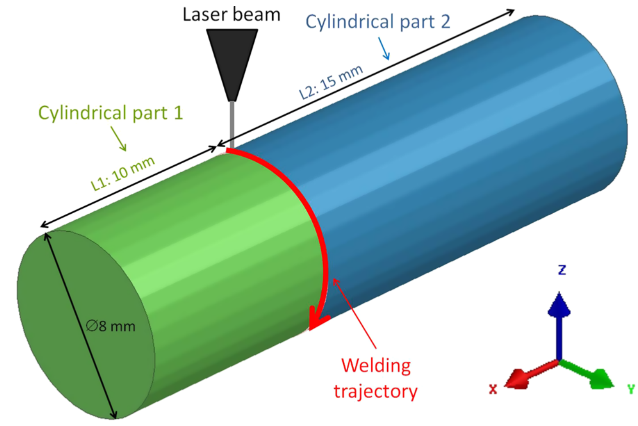


Fig. 1—Geometry used for the simulation of laser keyhole welding of dissimilar metals.

between the two parts that form the final component when defining the welding strategies (displacement and tilting of laser beam), as is explained in paragraph 2.2.4.

The two cylindrical parts that form the final revolution component are made of dissimilar materials. Specifically, the combinations of materials studied are:

- AISI 304 (X5CrNi1810) austenitic stainless steel and AISI 420 (X20Cr13) martensitic stainless steel
- AISI 316L (X2CrNiMo19-12) high-alloyed austenitic steel and AISI 1035 (CF35) carbon steel
- AISI 316L (X2CrNiMo19-12) high-alloyed austenitic steel and Inconel 718

All the materials used in the simulations have been selected among those included in the material database of SYSWELD®, the FEM (finite element method) software used for the simulations. The selection of combination of materials has been performed taking also into account real industrial applications: joining of ferritic or martensitic to austenitic stainless steels and ferritic to martensitic steels are very common in nuclear power generation industry,[4,6,7,40–42] in the chemical, petrochemical, and oil and gas industries,[4,6,7] especially in heat exchangers[43] and hydraulic valves,[20,21] and in the automotive industry[44]; joining of nickel-based alloys to alloy steels is commonly used in turbocompressor rotors[45] and in power plant and electrical applications.[42]

B. *FEM Simulation of Laser Keyhole Welding*

For the simulation of laser keyhole welding, the FEM commercial software SYSWELD® has been employed.

1. *Mesh*

The mesh model of the bimetallic cylindrical component used in the simulations is presented in Figure 2. The model contains more than 25,000 nodes, forming both tetrahedral and hexahedral elements. A finer mesh has been created in the joint area (Figure 2(b)), with element sizes of around 0.025 mm, in order to capture the steep temperature gradients with higher accuracy.

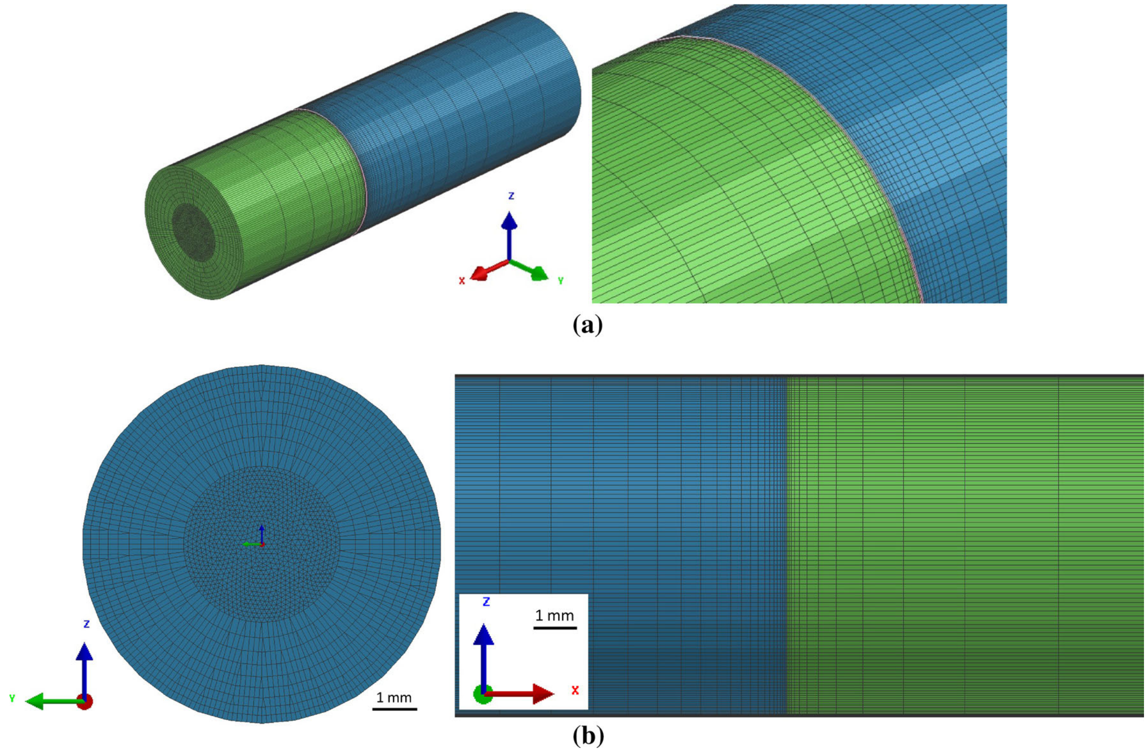


Fig. 2—FEM mesh of the bimetallic cylindrical component used in the simulations. (*a*) Isometric views of the whole component (left) and of a detail of the joint area (right). (*b*) Details of the mesh at the joint area (welding zone of the two cylindrical parts) in YZ (left) and XZ (right) views.

2. *Heat source*

In laser welding, the laser energy is absorbed by the base metal and converted into heat. The process of laser deep penetration welding (high depth-to-width ratio) is characterized by the existence of plasma and a keyhole. The keyhole cavity is kept open by the pressure of the plasma formed by ionized metal vapor.[46] This pressure acts continuously against the surface tension, which favors contraction. When quasi-steady-state conditions are reached, the shape of the keyhole and melt pool is established.

From finite element point of view, the heat source (laser beam) is modeled in SYSWELD® by a volume density of energy, Q (W/mm^3), applied to elements; this volume density of energy moves along the welding trajectory. Choosing a right heat source is very important in welding simulation in order to describe correctly the physical phenomena occurring during the welding process.[25] In the computation of laser keyhole welding process, the laser beam is commonly represented by a three-dimensional conical Gaussian heat source (Figure 3). At any plane perpendicular to the z-axis, the heat intensity distribution (Gaussian form) may be written as

$$Q_z = Q_0 \exp\left(-r^2/r_0^2\right), \qquad [1]$$

where Q_0 is the maximum heat intensity, r_0 is the radius of the heat source at z, and r is the radial coordinate of the interior point. The height of the conical heat source is $H = z_e - z_i$, z_e and z_i are being the heights of the top and bottom surfaces, respectively. The magnitudes r and r_0 are given as

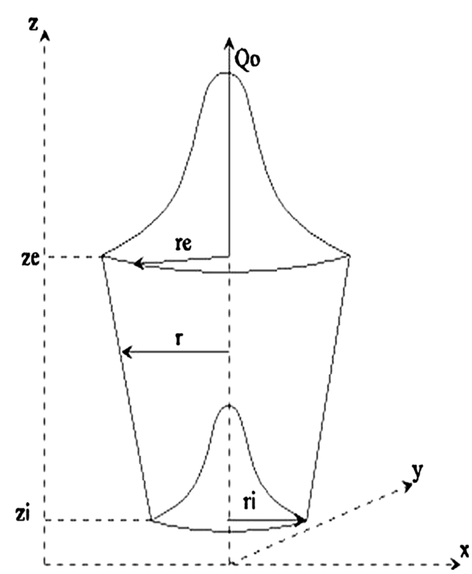


Fig. 3—3D conical heat source for keyhole welding modeling.

$$r^2 = \left(x^2 + y^2\right)$$
$$r_0^2 = r_e - (r_e - r_i)\frac{z_e - z}{z_e - z_i}, \qquad [2]$$

r_e and r_i are the radii at the top and bottom surfaces, respectively.

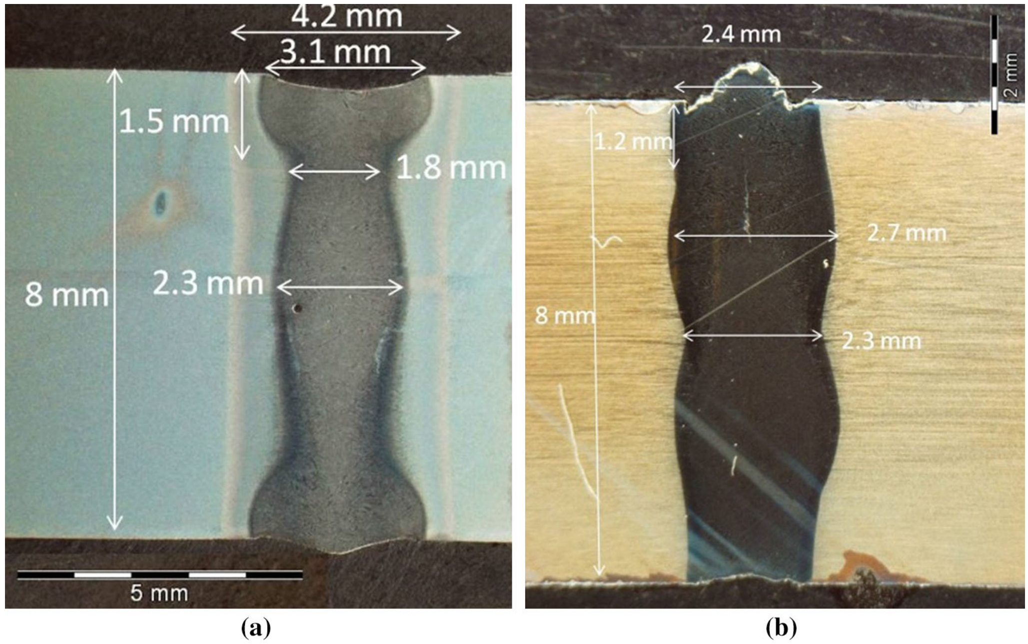


Fig. 4—Experimental longitudinal cross section of laser welding of (*a*) martensitic stainless steel and (*b*) austenitic stainless steel (non dissimilar welding but welding of each steel with itself).

3. *Heat input fitting*

The initial step for the simulation of the welding process in SYSWELD® FEM software is to adjust the parameters of the heat source (in the case of keyhole welding, the 3D conical heat source defined in the preceding paragraph). To do this, the module Heat Input Fitting, specific to this end, within SYSWELD® simulation software has been employed.

Initial experimental tests of welding of austenitic and martensitic stainless steels (non dissimilar welding but welding of each steel with itself) proved that full welding penetration can be obtained in both steels with a laser power $P = 1800$ W and a welding speed of 14.33 mm/s, as can be seen in Figure 4. Iterative simulations have been done in the Heat Input Fitting module of SYSWELD® varying r_e, r_i, z_e, and z_i parameters of the model heat source (Figure 3; Eqs. [1] and [2]) until the dimensions of the heat-affected zone (HAZ) and the molten area obtained from the simulation correspond to those observed experimentally (Figure 4). In the simulations, to determine the molten zone and the HAZ a graphical representation of the maximum temperature values reached is used: the molten area corresponds to the area of the material where temperatures higher than the liquidus temperature are reached (above liquidus temperature, metal will be melt); the HAZ corresponds with those areas where the temperature of the material surpasses 1073.15 K (800 °C). 1713.15 K (1440 °C) is the liquidus temperature of the austenitic steel and 1773.15 K (1500 °C) is the liquidus temperature of the martensitic steel, so these temperatures were used to set the limits of weld pool in each steel.

4. *Simulation models and parameters*

The model used in SYSWELD® FEM software for computing metallurgical transformations is based on the Leblond equation for diffusion controlled transformation kinetics:

$$\frac{\mathrm{d}P(T)}{\mathrm{d}t} = f(\dot{T})\frac{P_{\mathrm{eq}}(T) - P(T)}{\tau(T)}, \qquad [3]$$

where T is temperature, P is the metallurgical phase proportion at time t, P_{eq} is phase proportion at equilibrium, and τ is a "time delay." Martensitic transformations follow Koistinen–Marburger kinetics:

$$m(T) = 1 - \exp(-b(M_s - T)), \qquad [4]$$

where m is martensite proportion, T is temperature, b is Koistinen–Marburger coefficient, and M_s is martensite transformation start temperature. The influence of the thermal history on the mechanical history results simultaneously from variations of the mechanical properties with temperature and from thermal expansions/contractions. For mechanical calculation isotropic strain hardening model has been used for computing the elastoplastic behavior. SYSWELD® has a specific module for computing hardness, which uses Maynier formula for steels with carbon wt pct below 0.5 and Leslie formula if carbon wt pct is higher than 0.5. Hardness is determined from the chemical composition of the materials and the volume fractions of the constituents of the microstructure obtained in the thermo-metallurgical simulation. A detailed description of all these models is gathered in SYSWELD® Reference Manual.

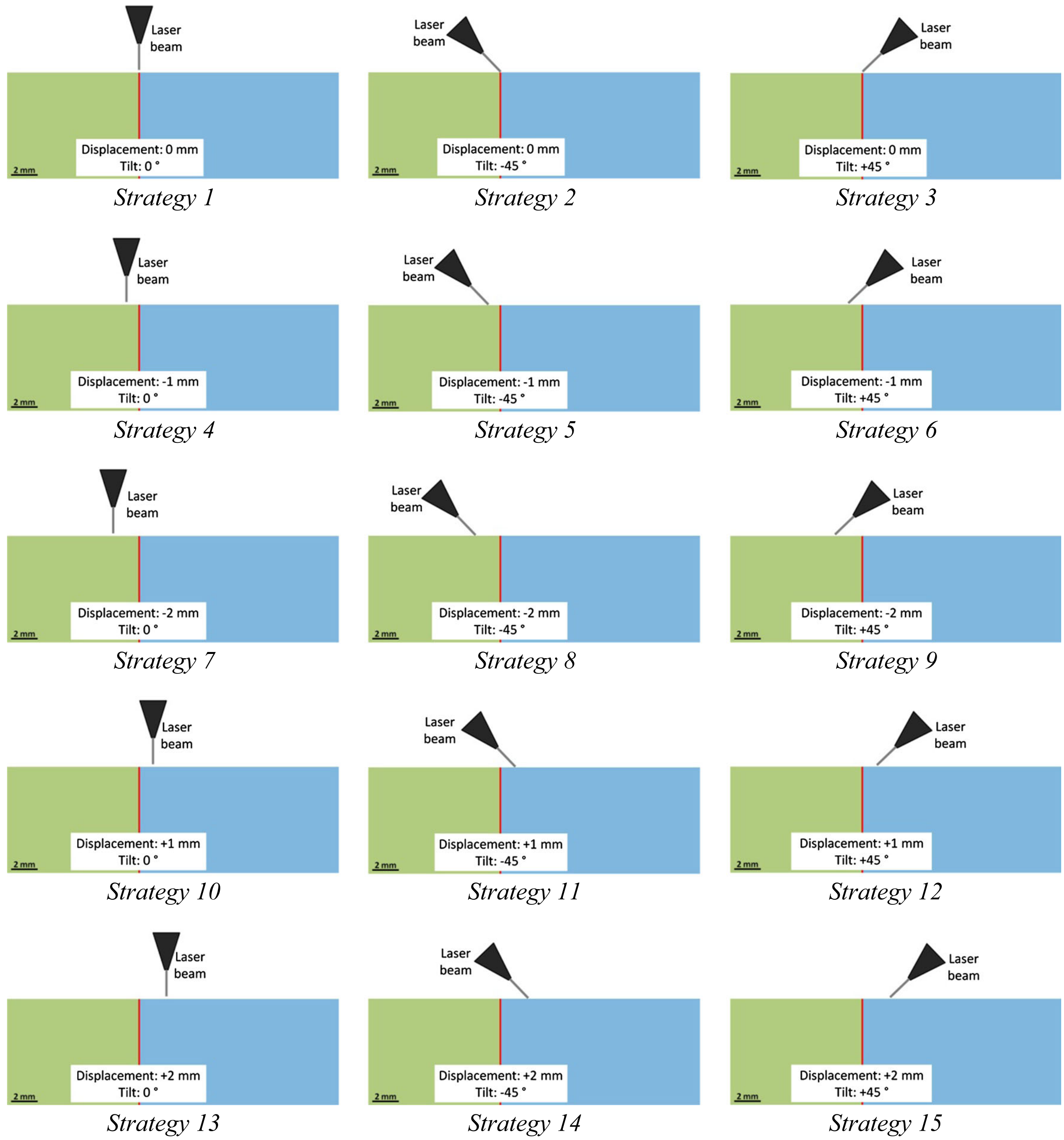


Fig. 5—Welding strategies (displacements and tilting of the laser beam) used in the simulations of laser keyhole welding of dissimilar metals.

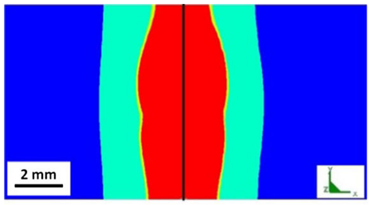
To completely define the model for laser keyhole welding simulation, the process parameters and the trajectory must be established. Regarding parameters, power as well as rotational speed are those used experimentally (laser power $P = 1800$ W and welding speed of 14.33 mm/s). Regarding the trajectory, a whole revolution along the joint has been considered. Perfect contact between the two cylindrical parts is assumed. Thermal boundary conditions used are thermal losses by radiation and convention. Three nodes of the base of one of the cylindrical parts have been used for clamping in order to avoid rigid solid movement of the component during the simulations.

5. *Welding strategies*

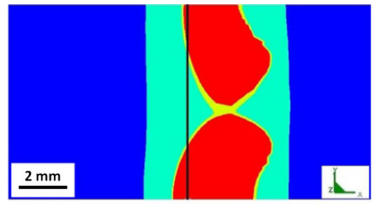
To evaluate the effect of displacing and/or tilting the laser beam (heat source), 15 different cases of simulation have been performed in this work (Figure 5): 5 different positions of the heat source (at the joint, displaced 1 mm toward the short cylindrical part, displaced 2 mm toward the short cylindrical part, displaced 1 mm toward the long cylindrical part, and displaced 2 mm toward the long cylindrical part) combined with 3 different tilting of the heat source (perpendicular, tilted 45 deg toward the short cylindrical part, and tilted 45 deg toward the long cylindrical part).

C. *Experimental Characterization of a Dissimilar Laser Keyhole Weld for Simulation Validation*

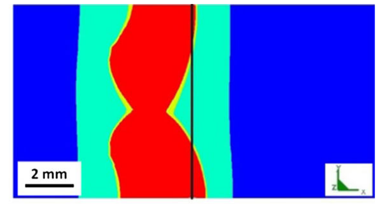
In order to validate the simulations, the microstructure, hardness, and residual stresses in a laser keyhole weld of austenitic to martensitic steel, with the laser



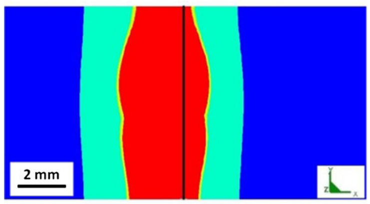
Strategy 1: Heat source perpendicular to the joint



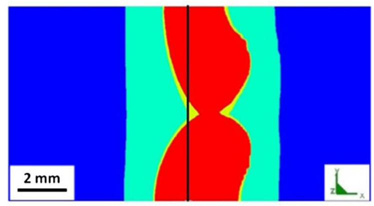
Strategy 2: Heat source tilted 45º towards the left of the joint (austenitic steel)



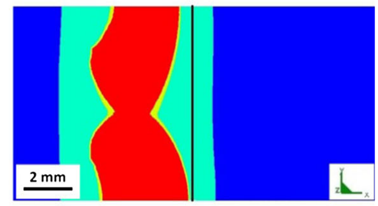
Strategy 3: Heat source tilted 45º towards the right of the joint (martensitic steel)



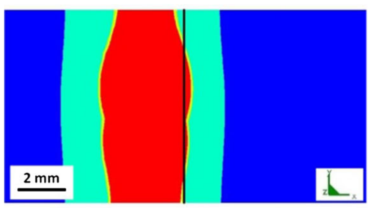
Strategy 4: Heat source moved 1 mm towards the left of the joint (austenitic steel)



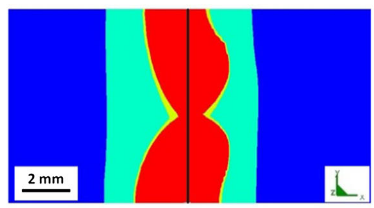
Strategy 5: Heat source tilted 45º and moved 1 mm towards the left of the joint (austenitic steel)



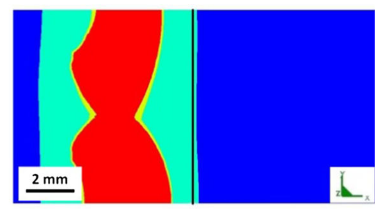
Strategy 6: Heat source tilted 45º towards the right of the joint (martensitic steel) and moved 1 mm towards the left of the joint (austenitic steel)



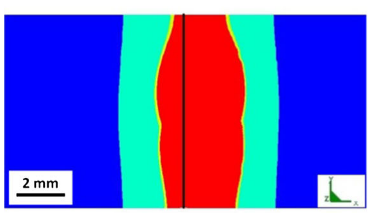
Strategy 7: Heat source moved 2 mm towards the left of the joint (austenitic steel)



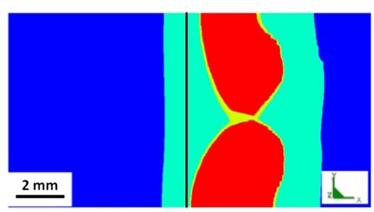
Strategy 8: Heat source tilted 45º and moved 2 mm towards the left of the joint (austenitic steel)



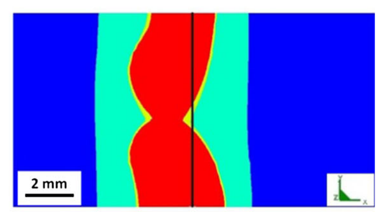
Strategy 9: Heat source tilted 45º towards the right of the joint (martensitic steel) and moved 2 mm towards the left of the joint (austenitic steel)



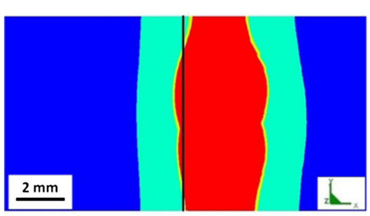
Strategy 10: Heat source moved 1 mm towards the right of the joint (martensitic steel)



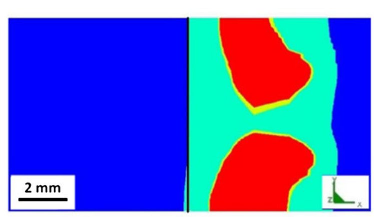
Strategy 11: Heat source tilted 45º towards the left of the joint (austenitic steel) and moved 1 mm towards the right of the joint (martensitic steel)



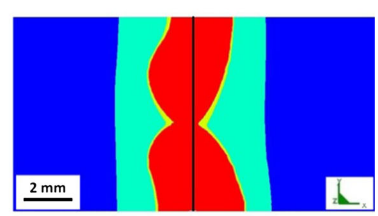
Strategy 12: Heat source tilted 45º and moved 1 mm towards the right of the joint (martensitic steel)



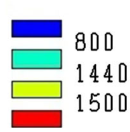
Strategy 13: Heat source moved 2 mm towards the right of the joint (martensitic steel)



Strategy 14: Heat source tilted 45º towards the left (austenitic steel) and moved 2 mm towards the right of the joint (martensitic steel)



Strategy 15: Heat source tilted 45º and moved 2 mm towards the right of the joint (martensitic steel)



Color scale of T_{max} (ºC)

Fig. 6—Longitudinal (XY) cross section at the joint area of the welded component indicating maximum temperatures ($T_{\max}$, °C) obtained during simulation of welding of AISI 304 (X5CrNi1810) austenitic stainless steel (left of the joint)and AISI 420 (X20Cr13) martensitic stainless steel (right of the joint). The joint is marked with a vertical black line (Color figure online).

beam perpendicular to the joint (welding strategy 1), have been analyzed. The laser employed has been a continuous Nd:YAG laser with $\lambda \sim 1064$ nm and maximum power of 2200 W. The laser is guided to the area of work through an optical fiber of circular cross section of 600 μm diameter. Focusing optical system is composed of a 200-mm collimator and a 200-mm focus lens. This system of lenses combined with the fiber system allows focusing the laser over the workpiece surface in a circular spot of 0.6 mm diameter. The laser head employed is a commercial welding head manufactured by Precitec, model YW50, which consists of two coaxial nozzles, so shielding gas flows coaxially with the laser beam, at 8 bar pressure.

For microstructural analysis, a polished and chemically etched (with Vilella's reagent) cross section of the laser weld has been observed in an optical microscope. Vickers hardness, using a load of 1 kg, has been measured in the same polished cross section used for microstructural study.

Surface residual stresses have been measured by means of X-ray diffraction using a Bruker D8 Advance diffractometer with parallel beam pollycap, PSD detector, and Cr radiation (wavelength $\lambda = 2.291$ Å). Measurement of surface residual stresses has been done using the $\sin^2 \psi$ method and following European Standard EN 15305:2008 (*Non-destructive Testing: Test Method for Residual Stress analysis by X-ray Diffraction*). (211) Fe-α diffraction peak located at $2\theta \sim 156$ deg and (220) Fe-γ diffraction peak located at $2\theta \sim 126$ deg have been used to measure surface residual stresses in the martensitic and austenitic steel, respectively. Nine ψ values from $\psi = -45$ deg to $\psi = +45$ deg at equispaced values of $\sin^2 \psi$ have been employed. To determine the X-ray diffraction peak position, Pearson VII adjustment of the peak profile has been employed, after applying the X-ray diffraction peak intensities corrections indicated in the European Standard EN 15305:2008, in particular absorption correction, background correction, $K_{\alpha 2}$ ratio correction, and Lorentz-Polarization correction. Error values in experimental surface residual stress values are provided by the analysis software employed (Leptos from Bruker), and these error values come from peak and d-$\sin^2 \psi$ fitting errors.

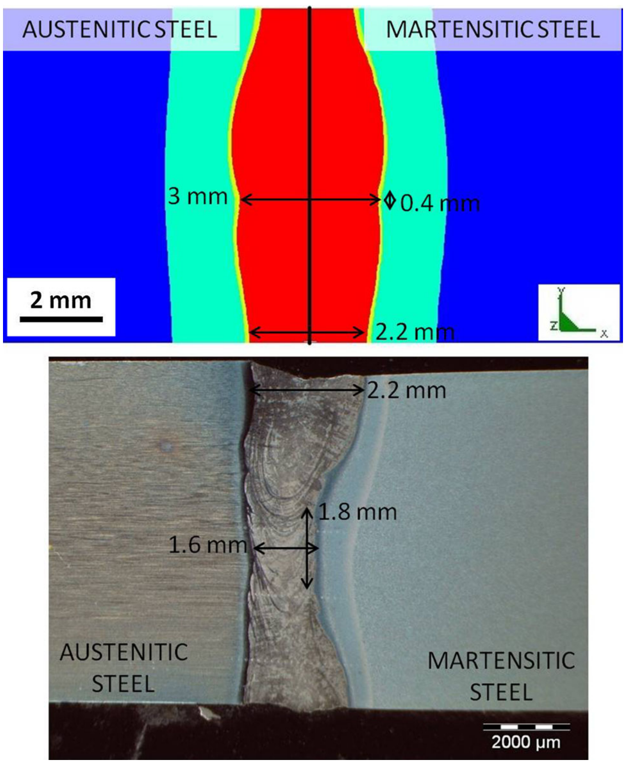


Fig. 7—Comparison of simulated (top) and experimental (bottom) longitudinal cross sections at the joint area of the welded joint of austenitic to martensitic steel with laser beam (heat source) perpendicular to the joint.

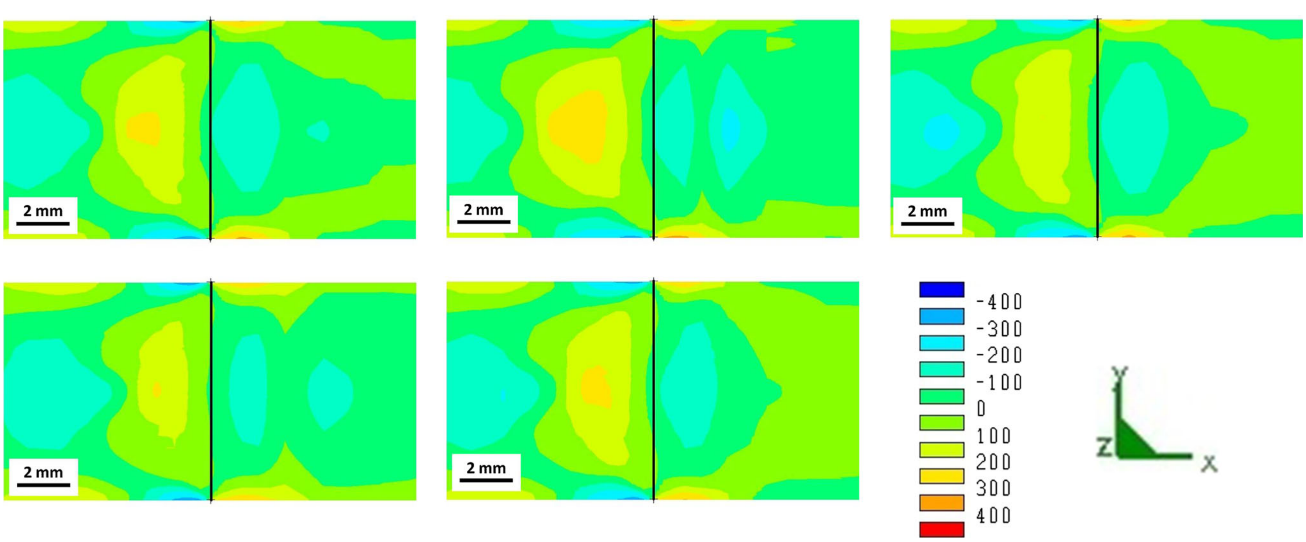


Fig. 8—Longitudinal (σ_{xx} component) residual stresses, in MPa, obtained by simulation, in a longitudinal cross section of the welded joint of AISI 304 (X5CrNi1810) austenitic stainless steel (left of the joint) and AISI 420 (X20Cr13) martensitic stainless steel (right of the joint). The joint is marked with a vertical black line (Color figure online).

III. RESULTS AND DISCUSSION

In all the simulations and experimental tests performed, the effect of welding is localized in the joint area and does not affect the rest of the long and short cylindrical parts that compose the component, so, from now on, the graphical representations will show only the joint area, *i.e.*, the localized area affected by the weld. In fact, this area around the weld bead and the heat-affected zone (HAZ) is the most critical region since in this area, the failures of welded joints during fatigue and cracking of dissimilar metal welds[47,48] are usually produced.

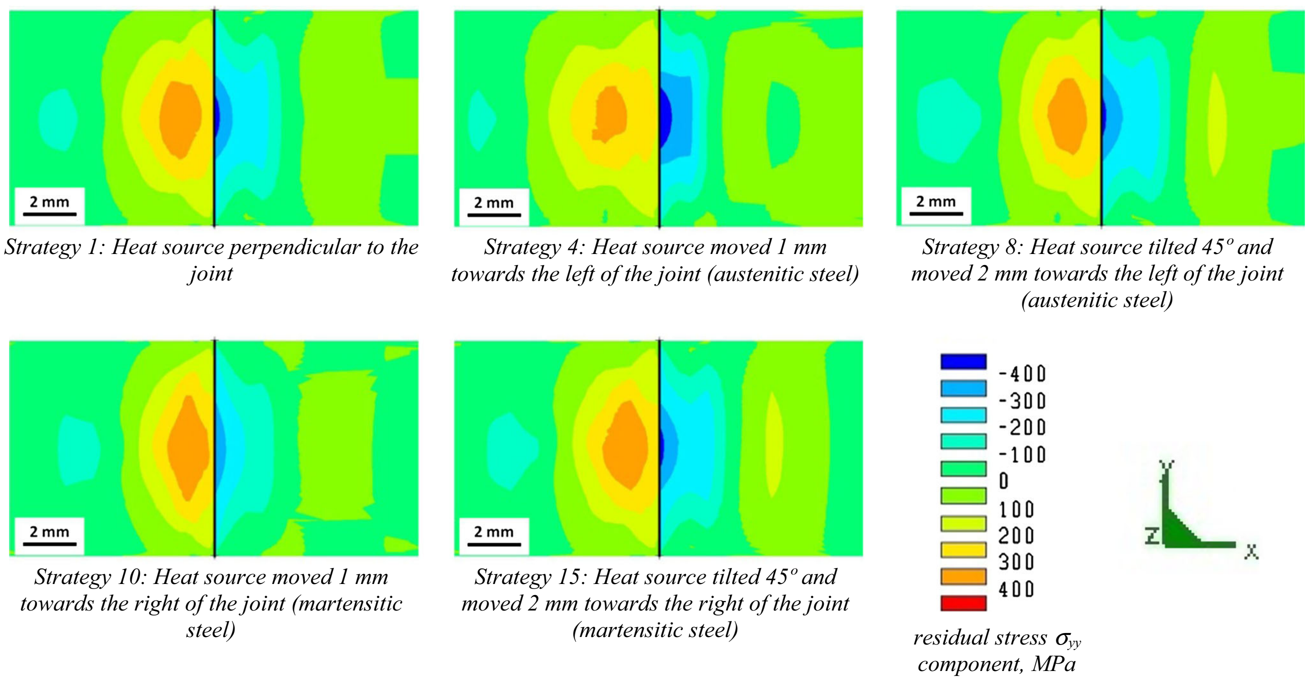


Fig. 9—Radial ($\sigma_{yy} = \sigma_{zz}$ component) residual stresses, in MPa, obtained by simulation, in a longitudinal cross section of the welded joint of AISI 304 (X5CrNi1810) austenitic stainless steel (left of the joint) and AISI 420 (X20Cr13) martensitic stainless steel (right of the joint). The joint is marked with a vertical black line (Color figure online).

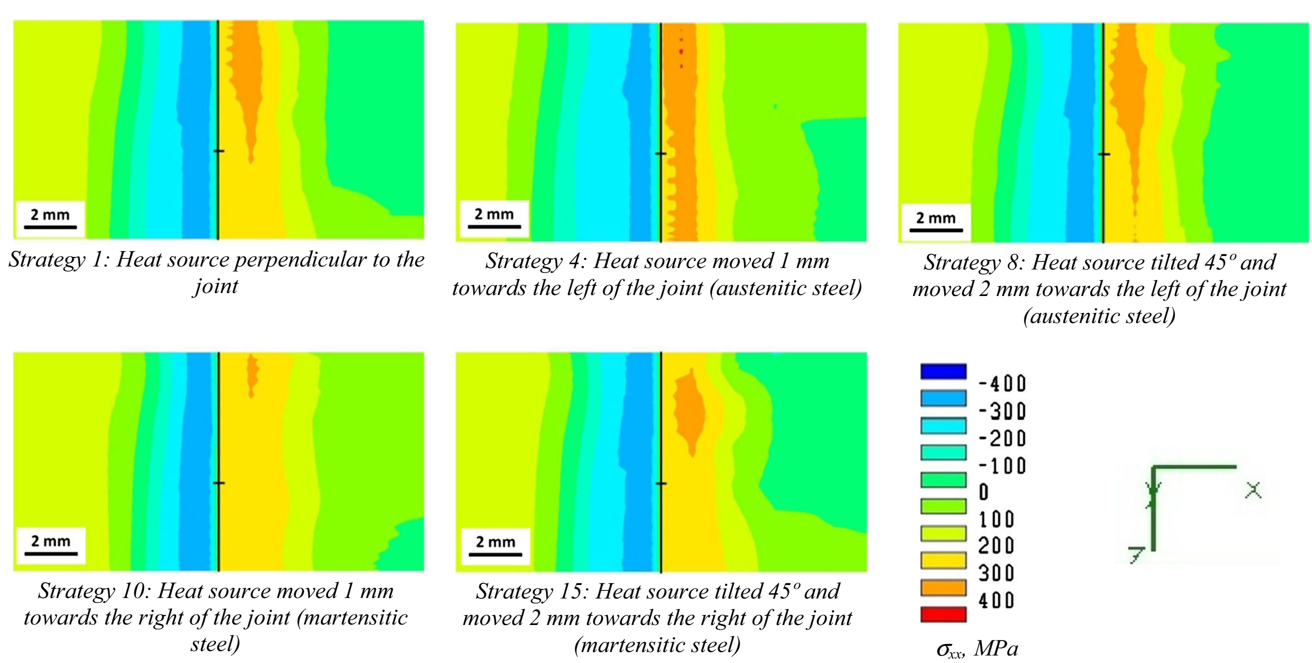


Fig. 10—Longitudinal (σ_{xx} component) residual stresses, in MPa, obtained by simulation, in the external surface of the welded joint of AISI 304 (X5CrNi1810) austenitic stainless steel (left of the joint) and AISI 420 (X20Cr13) martensitic stainless steel (right of the joint). The joint is marked with a vertical black line. The small horizontal black line indicates the start and end of laser trajectory (Color figure online).

A. *Laser Keyhole Welding of AISI 304 Austenitic Stainless Steel and AISI 420 Martensitic Stainless Steel*

Figure 6 gathers longitudinal cross sections of the joint area showing the maximum temperatures reached during simulation of welding of AISI 304 austenitic and AISI 420 martensitic stainless steels for the 15 different welding strategies. The temperature color levels used in this figure are dark blue for temperatures inferior to 1073.15 K (800 °C), *i.e.*, dark blue color indicates material not affected by the welding; light blue for 1713.15 K (1440 °C) > $T_{\max}$ > 1073.15 K (800 °C), which indicates the HAZ in both steels; yellow for 1773.15 K (1500 °C) > $T_{\max}$ > 1713.15 K (1440 °C) [1713.15 K (1440 °C) is the liquidus temperature of the austenitic steel, so $T_{\max}$ > 1713.15 K (1440 °C) indicates the limit of the weld pool in the austenitic steel]; and red for $T_{\max}$ > 1773.15 K (1500 °C) [1773.15 K (1500 °C) is the liquidus temperature of the martensitic steel, so $T_{\max}$ > 1773.15 K (1500 °C) indicates the limit of the weld pool in the martensitic steel].

Both steels must reach liquidus temperature along the entire joint (vertical black line in the pictures of Figure 6) in order to weld the two parts. As can be seen on Figure 6, fusion of both materials at the joint, *i.e.*, full penetration welding, is reached only in 5 of the 15 welding strategies studied: strategy 1 (Heat source perpendicular to the joint), strategy 4 (Heat source moved 1 mm toward the austenitic steel), strategy 8 (Heat source tilted 45 deg and moved 2 mm toward the austenitic steel), strategy 10 (Heat source moved 1 mm toward the martensitic steel), and strategy 15 (Heat source tilted 45 deg and moved 2 mm toward the martensitic steel).

In Figure 7, simulated and experimental longitudinal cross sections at the joint area of the welded joint of austenitic to martensitic steel with laser beam (heat source) perpendicular to the joint are compared. The dimensions of the weld pool (melt metal) in the surface of the component are identical, but there are two differences in the experimental and predicted weld pool below the surface, *i.e.*, in the bulk of the cylindrical component:

1. The symmetry: simulation predicts a symmetric weld pool at the joint, but experimental cross section shows a certain asymmetry in the molten metal.
2. The neck at the center of the joint: the weld pool is narrower at the center, but simulation predicts a broader but shorter neck than that observed experimentally (see dimensions indicated on Figure 7).

The shape of the welding pools obtained from simulations when displacing and tilting the laser beam are similar to those obtained experimentally by, for example, Li and Fontana[20] and Cantello *et al.*,[21] although exact dimensions of the weld pool cannot be compared because of the different materials and process parameters employed.

Regarding residual stresses, for all the welding strategies a change in the sign of stresses from the austenitic to the martensitic part is observed, *i.e.*, strong residual stress gradients are generated at the weld joint. Shear components (σ_{ij}, for $i \neq j$) are practically zero (values oscillate between −50 and +50 MPa) for all the welding strategies, while σ_{xx}, σ_{yy}, and σ_{zz} components are significant in magnitude and distribution. σ_{xx} is the stress component in the longitudinal direction of the cylindrical parts. σ_{yy} and σ_{zz} components are identical, as expected, since both components act in radial direction, and symmetry is expected in the results by the own symmetry of the part.

Longitudinal residual stresses, σ_{xx} (Figure 8), are practically identical both in values and distribution in the 5 welding strategies where full penetration is reached. The only difference is that the area of material inside the austenitic steel part with tensile stresses higher than 200 MPa is slightly higher in strategy 4 (Heat source moved 1 mm toward the austenitic steel) and slightly smaller in strategy 10 (Heat source moved 1 mm toward the martensitic steel). This small difference is probably negligible regarding its effect in the performance of the weld.

Radial residual stresses, $\sigma_{yy} = \sigma_{zz}$ (Figure 9), present similar values and distributions in simulation strategies 1, 8, and 15. Just at the center of the joint, in the martensitic steel part, higher compressive residual stresses and that extend over a higher area are generated in strategy 4 (heat source moved 1 mm toward the austenitic steel). In the bulk of the austenitic steel part, an area of material with tensile stresses higher than 300 MPa is generated; that area is higher and nearer to the joint in strategy 10 (heat source moved 1 mm toward the martensitic steel) than in the other strategies, and smaller and located at greater distance from the joint in strategy 4 (heat source moved 1 mm toward the austenitic steel).

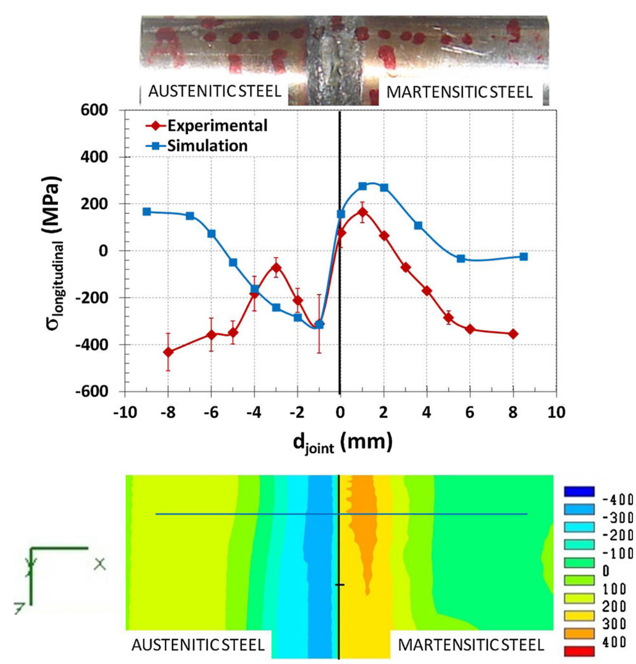


Fig. 11—Simulated and experimental (X-ray diffraction measurement) longitudinal (σ_{xx} component) residual stresses, in MPa, as a function of distance to joint, d_{joint} (mm), in the external surface of the welded joint of austenitic to martensitic steel with laser beam (heat source) perpendicular to the joint (Color figure online).

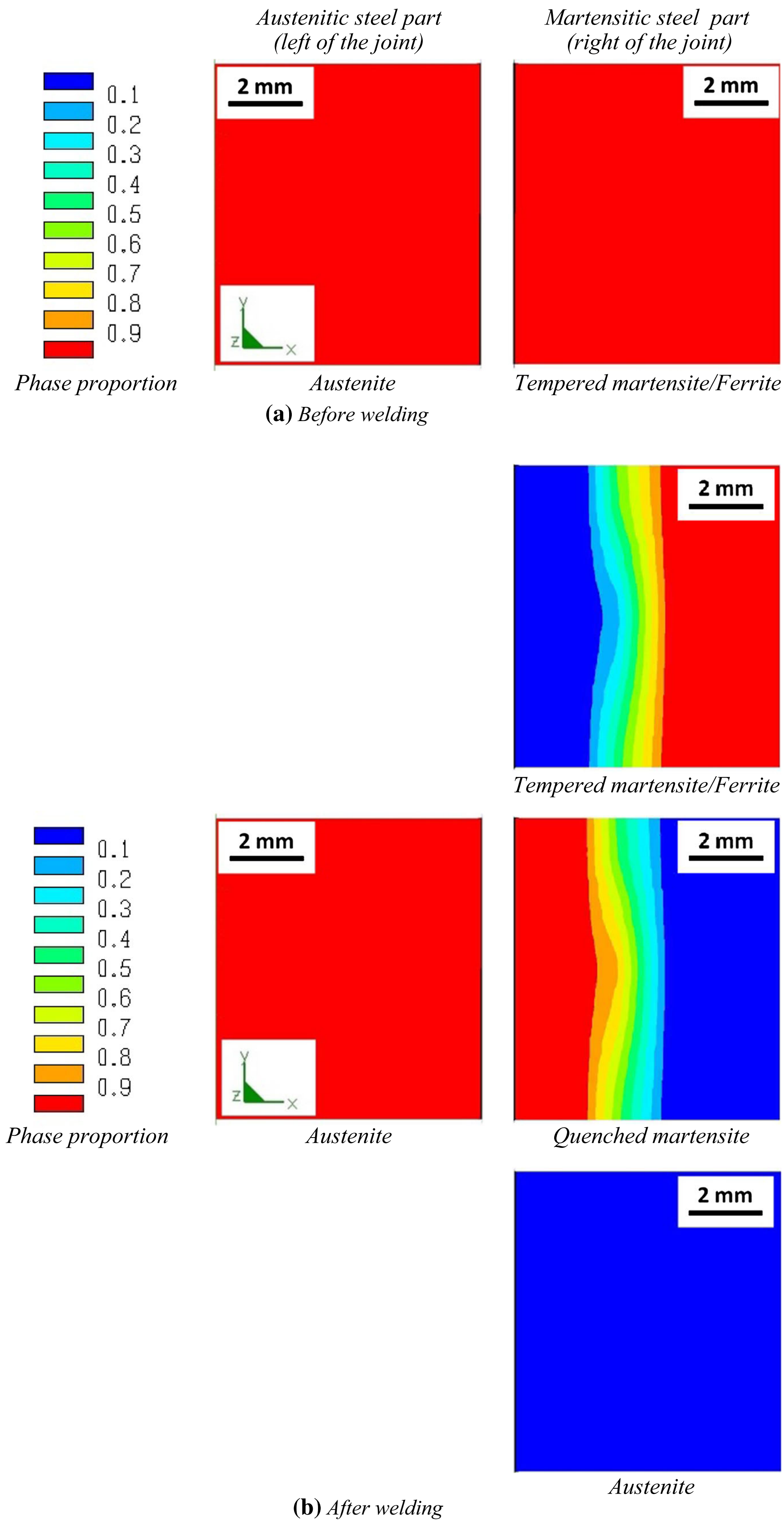


Fig. 12—Color maps of the phase proportions, (*a*) before and (*b*) after simulation of welding, in a longitudinal cross section of a welded joint of austenitic to martensitic steel with laser beam (heat source) perpendicular to the joint (Color figure online).

Therefore, the residual stress distribution generated when heat source is moved 1 mm toward the austenitic steel part (strategy 4) seems the most favorable. This agrees with experimental observations of Li and Fontana[20] and Cantello *et al.*[21] that concluded that a certain offset of the laser beam toward the austenitic part permits to control the melting ratio of an austenitic-ferritic steel joint, avoiding solidification cracking in the fusion zone. They also found that an impingement angle of the laser beam (plus the offset toward the austenitic part) helps avoiding also micro-fissuring in the HAZ.

All the components of stress tensor present maximum values at the center of the cylindrical parts in the joint area except σ_{xx} component: σ_{xx} maximum values are

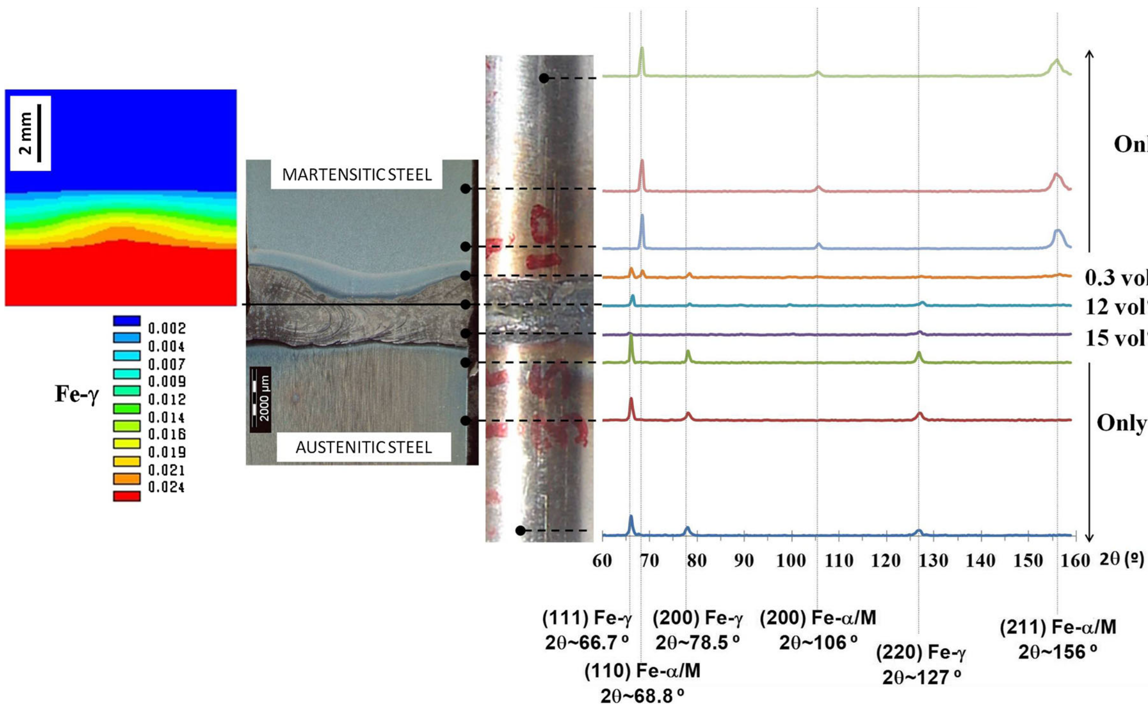


Fig. 13—Comparison of (left) simulated phase changes (simulated austenite proportion) after welding, (center) experimental optical cross section showing microstructure, and (right) experimental phases measured by X-ray diffraction at the surface of the welded joint of austenitic to martensitic steel with laser beam perpendicular to the joint (Fe-γ: austenite, Fe-α/M: martensite/ferrite) (Color figure online).

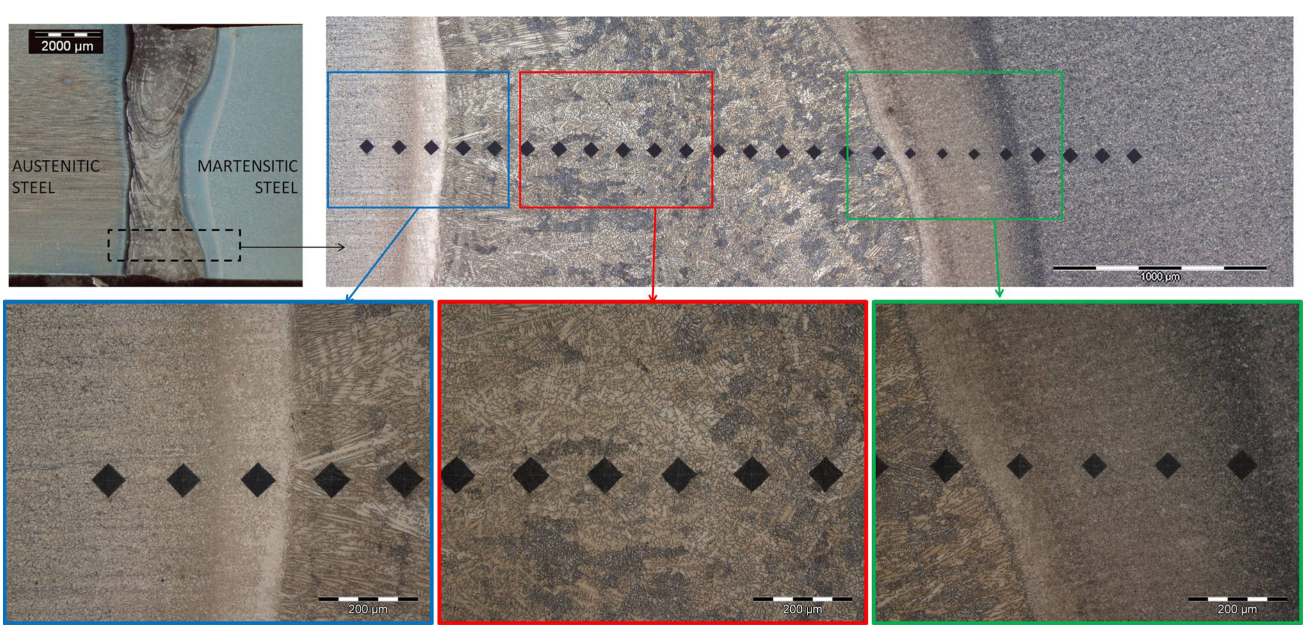


Fig. 14—Optical microscope images showing details of the microstructure in a longitudinal cross section of a welded joint of austenitic to martensitic steel with laser beam (heat source) perpendicular to the joint (Color figure online).

reached at the surface of the components, as seen in Figure 8 (cross section) and more clearly in Figure 10, where final residual stress σ_{xx} component at the surface of the part in the joint area has been represented for the 5 cases where full penetration is reached.

In Figure 11, simulation and experimental surface longitudinal residual stresses for welding strategy 1 (heat source perpendicular to the joint) are compared. The main difference between experimental and simulated surface residual stress profiles is in the area far away the joint, *i.e.*, in the non-affected material, where the simulations indicate null or nearly null stresses, whereas experimentally compressive stresses of around −350 MPa have been measured. This discrepancy can be explained by the fact that in the simulations the parts have a null stress state prior to welding, whereas experimentally, the welded cylindrical parts have a surface compressive stress state resultant of grinding of the parts: ground steel bars present surface longitudinal residual stresses prior to welding of −366 ± 50 MPa the austenitic steel bars and of −331 ± 40 MPa the martensitic steel bars (experimentally measured values). In the joint area, where the temperatures reached during welding are high enough to remove the previous residual

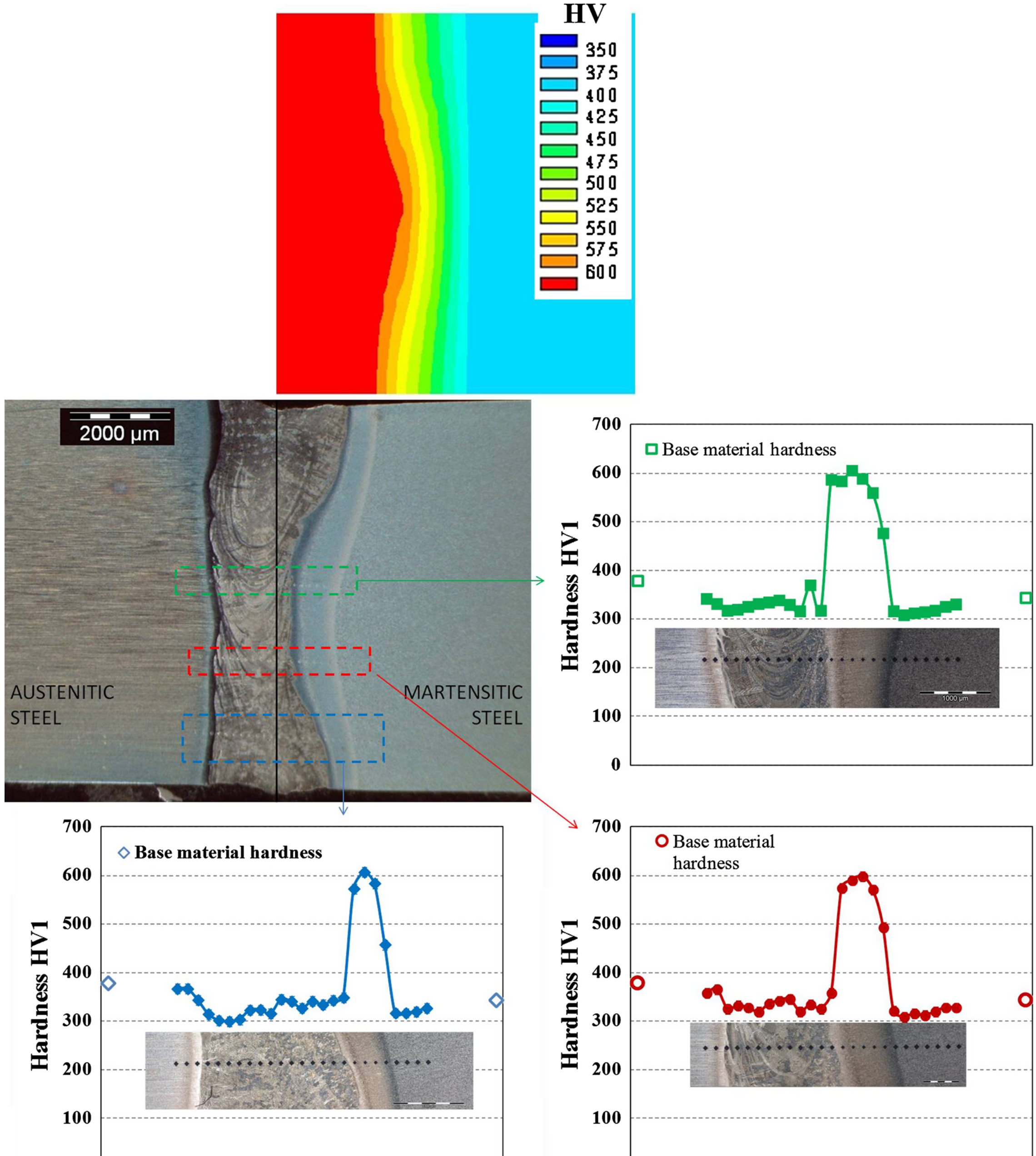


Fig. 15—Simulation (color contour maps of HV, top) and experimental (bottom) hardness values measured along three different lines crossing the joint in a longitudinal cross section (optical microscopy experimental image) of a laser-welded joint of austenitic and martensitic steels, with laser beam perpendicular to the joint (Color figure online).

stress state, the simulations reproduce exactly the stress profiles measured experimentally, so it can be concluded that the simulations are validated by the experimental results.

In steel welds, the transformation strains due to solid state phase transformation such as austenite to martensite or ferrite during cooling can strongly influence the residual stress.[38] Compressive stresses in the core of the joint area at martensitic steel (Figures 8 and 9), can be associated with the formation of martensite during the process, since martensite formation is commonly related with generation of compressive stresses due to the volumetric changes intrinsic to this phase transformation.

In Figure 12, distribution of phases have been represented before and after the simulation of welding with heat source perpendicular to the joint (welding strategy 1). Before welding, the two cylindrical parts are composed of 100 pct phase 1 (initial phase) in the simulation program, but this initial phase is different in each part: initial phase in the austenitic steel part is austenite and initial phase in the martensitic steel part is tempered martensite/ferrite. After welding, the austenitic steel part remains 100 pct austenite (initial phase); in the martensitic steel part, in the area near the junction, there is a gradual phase change: approximately 90 pct of quenched martensite (resulting from the melting and rapid cooling after the welding process) and 10 pct of tempered martensite/ferrite (initial phase) just next to the joint that gradually transforms to the 100 pct of initial tempered martensite/ferrite in the non-affected area of this part.

In Figure 12, the amount of austenite in the martensitic steel part after welding seems to be 0 due to the color scale used, but it must be pointed out that the simulation program provides a maximum value of austenite in the martensitic steel part of 0.024 (*i.e.* 2.4 vol pct). In Figure 13 (right), are presented X-ray diffraction patterns experimentally measured at different points of the surface of a sample welded with the heat source perpendicular to the joint. In Figure 14, some optical microscope images are gathered, showing details of the microstructure in the joint area of austenitic to martensitic steel welded experimentally. In these figures, and specially in Figure 13, the mixture of materials in the fusion zone is clearly seen: fusion zone is a mixture of austenite (Fe-γ) and Ferrite/Martensite (Fe-α). The austenite vol pct values presented in Figure 13 next to the X-ray diffraction patterns have been calculated using the intensities of 3 diffraction peaks of austenite and 3 diffraction peaks of martensite, according to ASTM International Standard E975-03: *Standard Practice for X-Ray Determination of Retained Austenite in Steel with Near Random Crystallographic Orientation*. In Figure 13 (left), the austenite phase proportion values obtained from simulation (cross section with automatic color scale) are also presented to compare them with the experimental results. Simulations reproduce the experimental ~0.3 vol pct austenite in the area next to the melt metal (weld pool) but the model considers a flat interface between the two materials and does not predict the mixing that occurs in the fusion zone, where experimentally measured volume fraction of austenite is around 15 vol pct due to the mixing of both austenitic and martensitic steels. The absence of metal mixing in the simulations is one of the main limitations of the model.

Phase changes measured experimentally and obtained from simulation must be reflected also in hardness changes. In Figure 15, experimental hardness profiles are presented. This hardness profiles have been measured along three different lines crossing the joint in a longitudinal cross section of a laser-welded joint of austenitic and martensitic steels, with laser beam perpendicular to the joint. Hardness of the molten metal (mixing of both steels) is around 330 HV1, lower than the hardness of both steels prior to welding: austenitic steel initial hardness was 379 HV1 and martensitic stainless steel initial hardness was 344 HV1 (experimentally

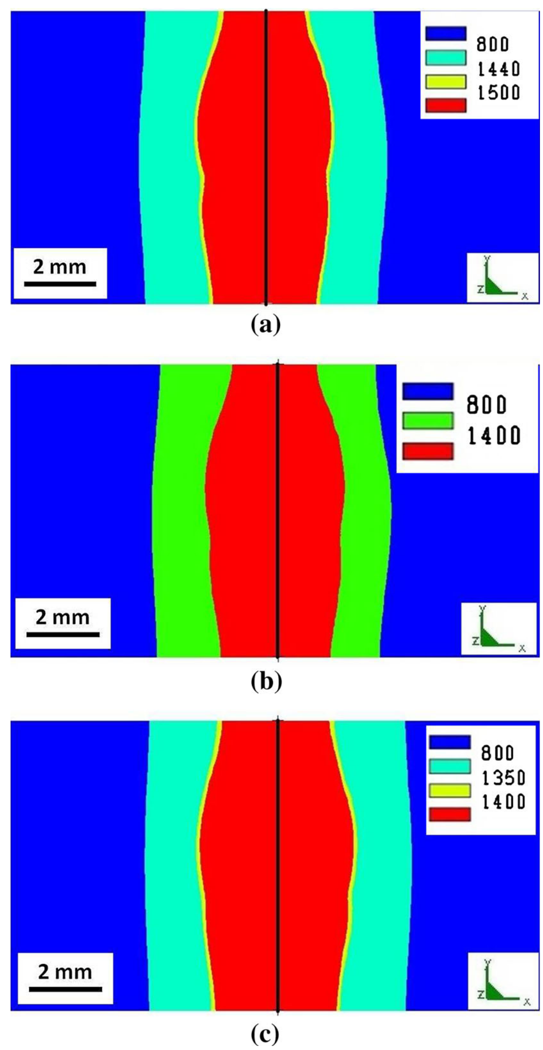


Fig. 16—Longitudinal cross section at the joint area indicating maximum temperatures (T_{max}, °C), obtained from simulation, reached during welding of (*a*) AISI 304 austenitic stainless steel (left of the joint) and AISI 420 martensitic stainless steel (right of the joint); (*b*) AISI 316L high-alloyed austenitic steel (left of the joint) and AISI 1035 carbon steel (right of the joint); (*c*) AISI 316L high-alloyed austenitic steel (left of the joint) and Inconel 718 (right of the joint) (Color figure online).

measured values). Next to the molten metal, toward the austenitic steel, there is a gradual increase in hardness toward the base material hardness. This gradual hardness change is produced in around 0.4 mm. In the area next to the molten metal toward martensitic steel, there is a sudden increase in hardness from the 330 HV1 of the molten mixed metal up to 600 HV1 in the adjacent area (area that presents a different color in the micrographs (area highlighted in the green box of Figure 14)), and that corresponds to quenched martensite (higher hardness than tempered martensite). These hardness changes agree with results obtained by Çam *et al.*[49] in a study of laser beam welding of different steels: they observed an extreme hardness increase in the weld regions of ferritic joints while no hardness increase was observed in laser-welded austenitic steel joints.

A hardness map obtained from simulation is also presented in Figure 15 (top). At the austenitic steel part, where no phase transformations occur, there is not a hardness change, so no graphical representation is presented. In the area of the martensitic steel part nearest to the junction, an increase in hardness associated with phase change is observed (Figure 15 (top)). Simulations predict a hardness of around 640 HV at the joint that decreases gradually toward the non-affected material hardness. This hardness of 640 HV is similar to that measured experimentally in quenched martensite (Figure 15 (bottom)). The main difference between simulation and experimentation results lies again in the fact that simulations consider a flat interface without mixing of metals, and therefore, the real increase in hardness is not produced in the initial interface but in the area next to the melt and mixed metals.

B. *Laser Keyhole Welding of AISI 316L High-Alloyed Austenitic Steel with AISI 1035 Carbon Steel and With Inconel 718*

Welding of AISI 316L (X2CrNiMo19-12) high-alloyed austenitic steel with AISI 1035 (CF35) carbon steel and with Inconel 718 has been simulated in order to study other combinations of dissimilar metals, employed also in industry, but that differ more regarding physical and thermo-mechanical properties.

These simulations have been compared with the previous ones (welding of AISI 304 austenitic and AISI 420 martensitic stainless steels). For that comparison, only one of the simulation cases is presented, in particular the simulation case with heat source perpendicular to the joint (simulation case 1).

In Figure 16, color maps with the maximum temperatures reached in a cross section of the joint are

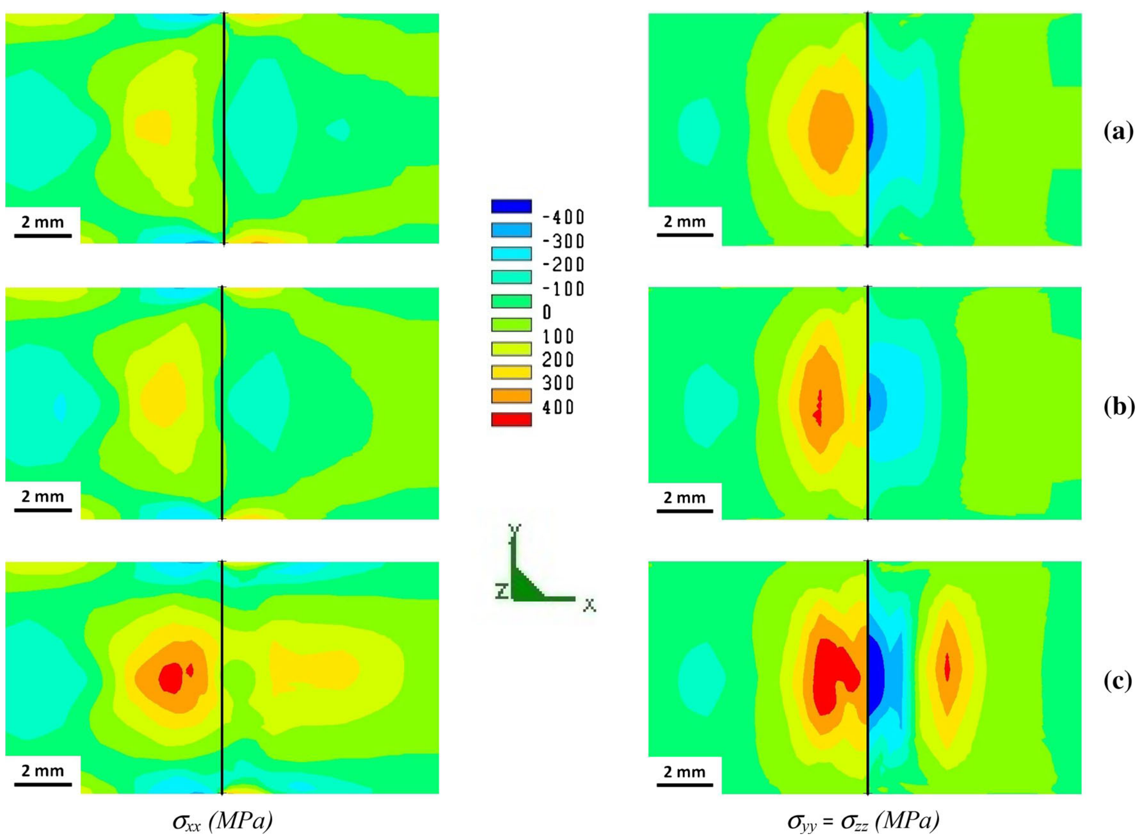


Fig. 17—Longitudinal (σ_{xx}) and radial ($\sigma_{yy} = \sigma_{zz}$) residual stresses, obtained by simulation, in a cross section of the joint area after welding of (*a*) AISI 304 austenitic stainless steel (left of the joint) and AISI 420 martensitic stainless steel (right of the joint); (*b*) AISI 316L high-alloyed austenitic steel (left of the joint) and AISI 1035 carbon steel (right of the joint); (*c*) AISI 316L high-alloyed austenitic steel (left of the joint) and Inconel 718 (right of the joint) (Color figure online).

gathered, indicating the heat-affected zone, $T_{max} > 1073.15$ K (800 °C), and the melt pool, $T_{max} > T_{liquidus}$ of each material [$T_{liquidus}$ AISI 304 = 1713.15 K (1440 °C); $T_{liquidus}$ AISI 420 = 1773.15 K (1500 °C); $T_{liquidus}$ AISI 316L = 1713.15 K (1400 °C); $T_{liquidus}$ AISI 1035 = 1713.15 K (1400 °C); and $T_{liquidus}$ Inconel 718 = 1623.15 K (1350 °C)]. No substantial differences are observed in the weld pool geometry and dimensions for the different dissimilar metals combinations.

In Figure 17, the σ_{xx} and $\sigma_{yy} = \sigma_{zz}$ components of residual stresses in the cross section of the joint for the three dissimilar welding simulations are presented. In austenitic to carbon steel joint (Figure 17(b)), the residual stresses are very similar to those obtained in the welding of austenitic to martensitic steel (Figure 17(a)), but welding of steel to Inconel 718 (Figure 17(c)) results in steeper residual stress gradients and higher area at the joint with detrimental tensile stresses. This indicates that when the difference in thermo-mechanical properties of the dissimilar metals to be welded is higher, the final stress state generated could be more detrimental to the performance of the component, and therefore, optimization of the welding strategy is more relevant. Material or material-related characteristics that influence the development of residual stress include, as Bhadeshia[50] states, thermal conductivity, heat capacity, thermal expansivity, elastic modulus and Poisson's ratio, plasticity, thermodynamics and kinetics of transformations, mechanisms of transformations, and transformation plasticity. In welding of dissimilar materials, the response of each material to the heat source will be different if material properties are different; for example, if the thermal expansion coefficients of the two metals differ greatly, there will be internal stresses set up in the intermetallic zone, associated with a thermal expansion mismatch. On the other hand, if each metal undergoes different phase transformations, residual stresses due to these phase transformations will be different, and also a global residual stress will be generated due to mismatch associated to volumetric changes generated in phase transformations.

In Figures 18 and 19, color maps of components of residual stresses in welding of austenite steel to Inconel 718 are presented, for the 5 different welding strategies where full penetration has been reached. As indicated in previous paragraph, in laser keyhole welding of austenitic to martensitic steel the residual stress distribution generated when heat source is moved 1 mm toward the austenitic short part (simulation strategy 4) seems the most favorable regarding final residual stresses, so it would be the best welding strategy (among those studied in present work). In the case of austenitic steel to Inconel 718 welding, this strategy is also the optimum regarding radial residual stresses (Figure 19), but in the case of longitudinal residual stresses (Figure 18), it seems the worst strategy, being the least detrimental the strategy where heat source is tilted 45 deg and moved 2 mm toward austenitic steel. Since residual stresses add up to the external stresses that the component suffers during service, depending on the direction of stresses suffered by the component resultant of the nominal load (load suffered by the component during its normal service) one strategy or the other would be the best regarding weld performance. In general, it can be said that residual stresses are beneficial when they operate in the plane of stresses generated by the applied load and are opposite in sense.

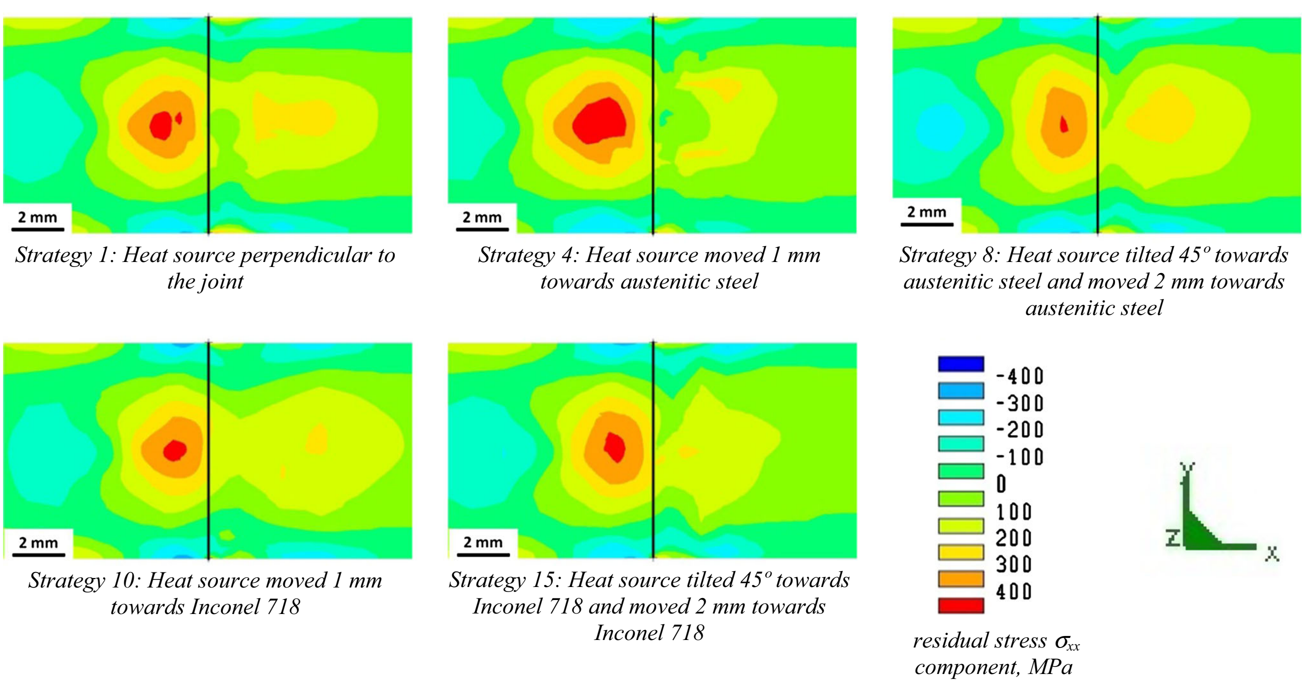


Fig. 18—Longitudinal (σ_{xx} component) residual stresses, in MPa, obtained by simulation, in a longitudinal cross section of the welded joint of AISI 316L high-alloyed austenitic steel (left of the joint) and Inconel 718 (right of the joint) (Color figure online).

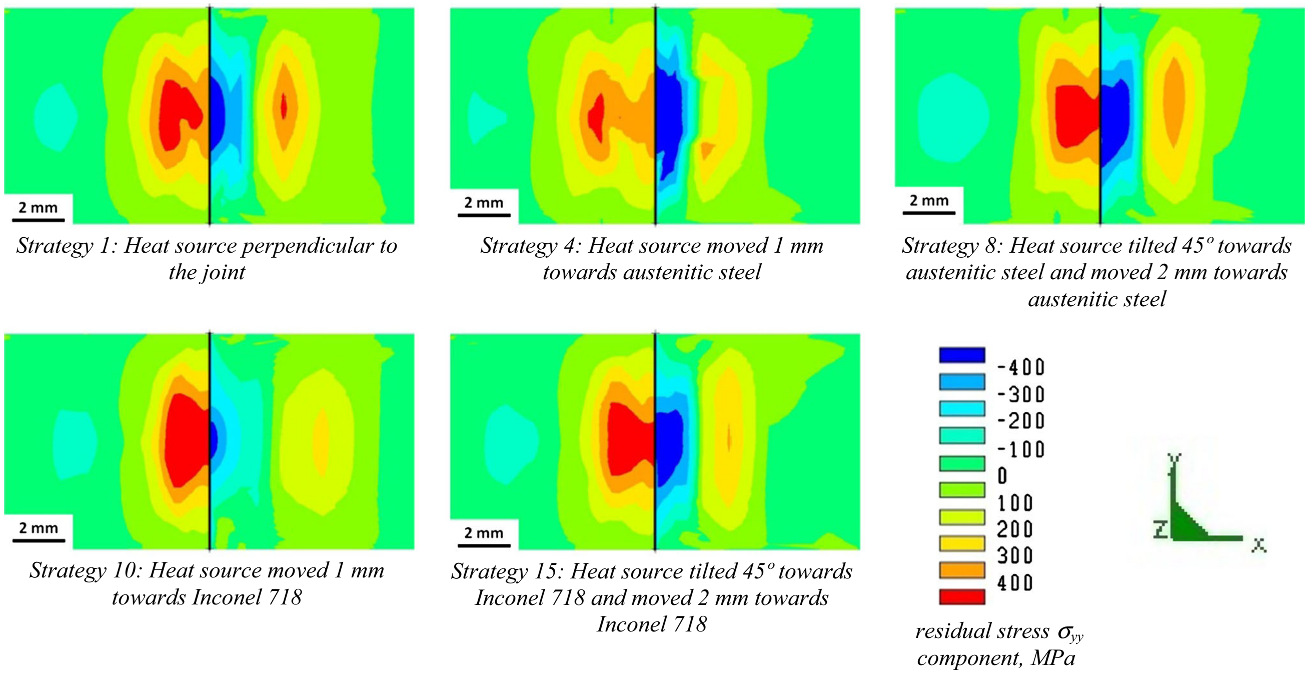


Fig. 19—Radial ($\sigma_{yy} = \sigma_{zz}$ component) residual stresses, in MPa, obtained by simulation, in a longitudinal cross section of the welded joint of AISI 316L high-alloyed austenitic steel (left of the joint) and Inconel 718 (right of the joint) (Color figure online).

IV. CONCLUSIONS

The simulation model used in this work has one important limitation: simulations consider a flat interface between the two materials and do not take into account the mixing of molten metals that occurs in the fusion zone. Outside this mixing area, simulations reproduce quite well experimental results (microstructure, hardness, and surface residual stresses after welding), so the model presented can be considered as experimentally validated, although with the limitation mentioned previously. Consequently, this model can be used to predict the behavior of the dissimilar laser welds if parameters of the process as laser power, speed, *etc*., and/or materials to be welded are modified, so it can be employed to optimize welding strategies, improving production of components and avoiding, at least in part, the costly experimental trial and error approach.

For a fixed welding strategy (heat source perpendicular to the joint), welding of different combinations of metals leads to similar weld pool geometries but very different residual stresses, so another important conclusion of this work is that weld pool geometry study is not enough to determine the quality of a joint: a complete study of residual stresses is fundamental since welds with similar weld pool geometries can present different distributions of residual stresses.

Laser keyhole welding with laser beam perpendicular to the joint of austenitic to carbon steel and austenitic to martensitic stainless steels results in similar residual stresses at the joint area, but welding of steel to Inconel 718 with the same strategy results in steeper residual stress gradients and higher area at the joint with detrimental tensile stresses. This indicates that when the difference in thermo-mechanical properties of the dissimilar metals to be welded is higher, the stress state generated is more detrimental for the service life of the component, and therefore, more relevant is the optimization of the welding strategy.

In the case of laser keyhole welding of austenitic to martensitic stainless steel and austenitic to carbon steel, the optimum welding strategy (less detrimental residual stresses at the joint area) consists in displacing the laser beam 1 mm toward the austenitic steel. In the case of austenitic steel to Inconel welding, the optimum welding strategy consists in having the heat source tilted 45 deg and moved 2 mm toward the austenitic steel.

ACKNOWLEDGMENTS

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